

DEVELOPMENT OF AN INTELLIGENT CHATBOT FOR STUDENT INFORMATION SUPPORT BASED ON LANGUAGE MODELS

STUDENT: AYAUZHAN BAURZHANOVA |
3RD YEAR STUDENT, SOFTWARE ENGINEERING

SUPERVISOR: ZHANNA SADIRMEKOVA

GITHUB: HTTPS://GITHUB.COM/AYAUZHAN-BAURZHANOVA/DIPLOMA-CHATBOT-WITH-MASCOT.GIT

YOUTRACK: HTTPS://AYAUB.YOUTRACK.CLOUD/AGILES/204-1/CURRENT

Problem Statement & Target Users

- Students waste time searching for specific rules across unstructured PDF documents
- Existing search lacks contextual understanding
- Information is scattered across Moodle sections, creating bureaucratic friction
- Administrative staff and advisors face a high volume of repetitive, trivial inquiries

Target users: current AITU students

Goals & Objectives

AIM:

To develop a student-centric, gamified intelligent mobile application, powered by a RAG architecture and a local SQLite database to deliver source-verified information regarding university regulations.

Objective 1: Aggregate and vectorize the university's knowledge base.

- Expected Result: A structured ChromaDB vector database containing fully parsed syllabuses and academic policies.

Objective 2: IIntegrate LLM via OpenRouter API with similarity thresholding and deterministic grade calculation.

- Expected Result: An asynchronous FastAPI backend answering academic queries and calculating GPA/FX scenarios

Objective 3: Develop an interactive UI featuring mascot animations.

- Expected Result: A deployed, highly engaging mobile client application built in Unity.

Objective 4: Execute rigorous accuracy, abstention, and latency testing against a 70-query benchmark dataset.

- Expected Result: Empirical validation achieving >85% extraction accuracy, >=95% out-of-scope abstention, and <8 seconds response latency.

Literature Review

Synthesis of 12 Sources (10 Scientific Publications & 2 Technical References)

Publication & System	Core Methodology / Stack	Key Findings	Limitation Addressed by RemoodleChat
Ong et al. (2026) (*AskNarelle*) & **Shihab (2026)**	**GPT-4o**, Azure AI Search (chunk=1500, overlap=150, top_k=10)	Reduces repetitive staff inquiries 24/7; "Reserved Followers" distrust unverified bots.	Lacks inline PDF section citations, grade math, and gamification; relies on paid Azure stack.
Joseph (2025) (*CoVe-RAG+*) & **Yuan et al. (2026)** (*SeaRAG*)	Iterative Chain-of-Verification + Explainability Layer; statement-entity ranking	Reduces closed-domain hallucinations by 28%; Explainability Layer builds user trust.	Multi-loop verification causes high latency; we adopt the Explainability Layer (strict citations) with single-pass thresholding ($\tau = 0.72$).
Sanyal et al. (2026) (*RAGMail*) & **Wu et al. (2025)** (*MultiRAG*)	**FastAPI, ChromaDB, LangChain**; multi-source authority metadata weighting	Improves factualness by 44.86% and resolves conflicting multi-document rules.	Multi-loop FEWL scoring and runtime graph building are slow; we apply authority & section tagging during offline PDF ETL (<8s latency).
Rafique et al. (2026) (*EduAssist*), **Niyazova et al. (2026)** & **Ryan & Deci (2020)**	Gamified AI environments (XP/Avatars) grounded in **Self-Determination Theory (SDT)**	85% usability & 70% motivation boost when rewards are optional and aligned with student utility.	Intrusive gamification harms utility; we link real assignment grades (1:1) to store points (P) with a collapsible UI & persona toggle.
Es et al. (2024) (*RAGAS*), **ChromaDB (2026)** & **Unity Mecanim (2026)**	Automated RAG metrics (Faithfulness, Relevance); HNSW vector indexing; 2D Animator state machines	Standardizes retrieval/generation QA; enables fast local metadata filtering & layered sprite swaps.	Integrated directly for Phase 4 verification (>85% accuracy on 70-query benchmark), sub-second chunk filtering, and emotion_id transitions.

Multi-pass verification RAG pipelines are too slow for real-time student chat. RemoodleChat combines single-pass similarity thresholding and inline citations with SDT-grounded, toggleable gamification to achieve speed and high student engagement.

ASTANA IT UNIVERSITY

Evidence of Relevance

Interview Responses

Key Themes (from 6 student-respondents, answering a 6-question interview):

Finding information on Moodle takes 5–20 minutes (4 of 6 respondents) due to downloading and scrolling through long PDFs.

- Shihab (2026) & Ong et al. (2026): Fragmented university portals delay student workflows and require automated retrieval.

Students requested hypothetical grade/scholarship calculations and have expressed zero tolerance for misinformation

- Joseph (2025) - RAG systems must include an "explainability layer" with source traceability to build user trust.

Must reply within 2 to 8 seconds.

- Aligns with Sanyal et al. (2026): Necessitates a lightweight, low-latency FastAPI + ChromaDB pipeline.

Gamified mascot and achievements are appealing, but need to be non-intrusive.

- Niyazova et al. (2026) & Ryan & Deci (2020): Gamification boosts motivation only when optional and aligned with core utility.

Verified Requirements: <8s latency, inline citations, deterministic grade math, non-intrusive UI; and Design
Hypothesis: A standalone Unity mascot app with a collapsible chat panel will achieve higher engagement than a plain Telegram bot.

Why RemoodleChat? A Comparative Analysis

Criteria	RemoodleChat (Proposed)	Standard ChatGPT	AITU / Moodle Search	Rule-Based FAQ Bot	AskNarelle (Ong et al., 2026)
1. AITU Local Rules & Syllabuses	**Complete** (Vectorized PDFs)	Low / Speculative	Fragmented (Manual PDFs)	Limited (Static Q&A)	Single-course only
2. Hallucination Prevention	**High** (RAG + τ = 0.72 cutoff)	Low (Invents rules)	N/A (Raw files)	High (Rigid tree)	High (RAG-grounded)
3. Inline Source Traceability	**Yes** (PDF Section & Page)	No	Partial (File link only)	Manual only	No section links
4. Semantic Understanding	**High** (Embeddings + LLM)	High	Low (Keyword only)	Low (Buttons/Keywords)	High (GPT-4o)
5. Grade & GPA Calculation	**Yes** (Deterministic tool)	Unreliable math	No	No	Escalates to human
6. Gamification & Mascot UI	**High** (Mascot, Store, Persona)	Low (Text only)	None	Low	Low (Web chat)
7. Retrieval Speed	**< 8 seconds**	~3–5 seconds	10–20 minutes	< 2 seconds	~4–8 seconds
8. Personalization	**High** (Gender & Outfit tone)	Manual prompting	None	None	N/A

Conclusion: Existing general LLMs hallucinate local rules, while university portals and academic RAG bots lack source-section traceability, deterministic grade calculation, and student-centric gamification. RemoodleChat bridges all three gaps in a single mobile ecosystem.

Derived System Requirements

- FR-1 (Must) — Multi-Source RAG & Conflict Resolution: Retrieve top-k (k=5) ChromaDB chunks (similarity>= 0.72), prioritizing official Academic Policy over syllabuses.
- FR-2 (Must) — Deterministic Scenario Calculation: Execute exact Python formula math for hypothetical RegMid/RegEnd/Final grade and scholarship queries (0% math error).
- FR-3 (Must) — Source Traceability: Provide a clickable reference link/icon to open the source document on 100% of factual bot answers.
- FR-4 (Must) — Onboarding & Personalization: Persist Name, Gender, Year (1st–3rd), and Faculty in SQLite to adapt forms of address.
- FR-5 (Must) — Convert manually logged assignment grades 1:1 into POINTS (P) in SQLite to buy/equip outfits.
- FR-6 (Should) — Dynamic Emotion & Outfit Persona (with Toggle): Trigger Unity Animator via emotion_id and shift persona by outfit (unless disabled in Settings).
- FR-7 (Should) — Persistent Chat History: Save and load past chat threads locally via SQLite.

Derived System Requirements

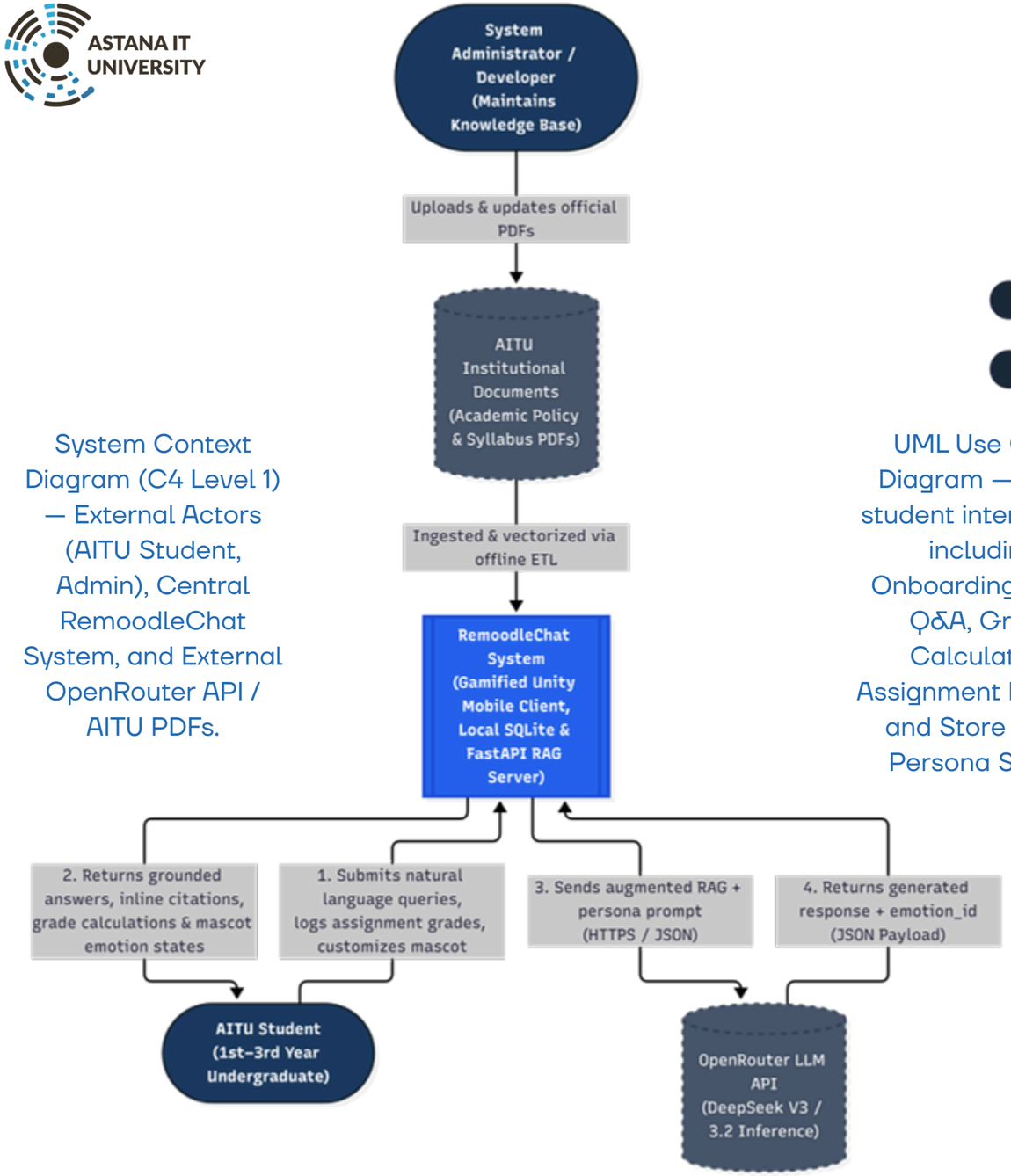
- NFR-1 (Must) — Hallucination Abstention: Achieve >85% in-scope accuracy and >=95% fallback warning activation when similarity < 0.72 on out-of-scope queries.
- NFR-2 (Must) — Low Latency: Mean end-to-end response time < 8.0 seconds (95th percentile < 10s).
- NFR-3 (Must) — Unobtrusive UI: Collapsing the mascot panel via the Enlarge icon expands text chat in < 200 ms.
- NFR-4 (Must) — Secret Isolation: Zero API keys in Unity binary or Git (.env isolated on FastAPI server).

Technology Selection Fulfilling FRs & NFRs

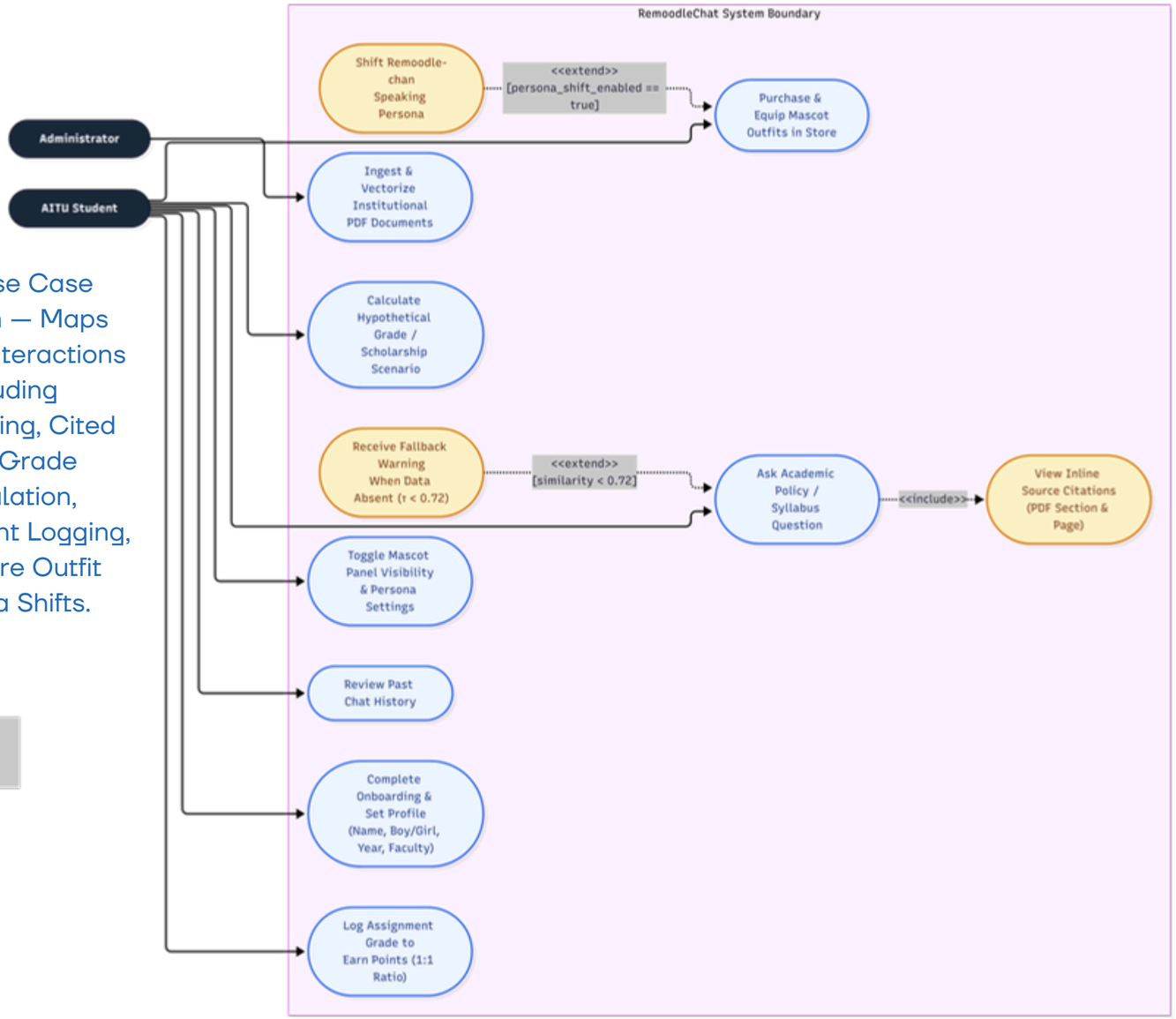
Subsystem	Selected Technology	Alternatives Considered	Justification & Requirement Alignment
1. Mobile Frontend & Animation	**Unity Engine (C#)**	Flutter (Dart), React Native, Native Android SDK	**FR-5, FR-6, NFR-3.** Native Animator state machine and 2D sprite layering handle outfit swaps and emotion_id transitions far better than standard UI toolkits.
2. Backend & Orchestration	**FastAPI (Python) + LangChain**	Django REST, Flask, Node.js (Express)	**FR-1, FR-2, NFR-2.** Native ASGI async/await concurrency and Python ML ecosystem integration keep retrieval under the <8s latency threshold.
3. Vector Database	**ChromaDB**	Pinecone, FAISS, Qdrant	**FR-1, FR-3, NFR-1.** Free, persistent local storage with native metadata filtering (section/page/authority) unlike paid Pinecone or raw FAISS.
4. Client Relational DB	**SQLite (sqlite-net-pcl)**	Unity PlayerPrefs, Remote PostgreSQL, Firebase	**FR-4, FR-5, FR-7.** Enforces relational integrity across Users, Assignments, StoreItems, UserInventory, and ChatMessages with zero network latency.
5. LLM Provider	**OpenRouter API (DeepSeek V3/3.2)**	OpenAI (GPT-4o), Direct DeepSeek API, Local Ollama	**FR-2, FR-6, NFR-2.** Low cost, high token speed, and hot-swappable routing without local hardware bottlenecks or direct API tariff spikes.

Architecture

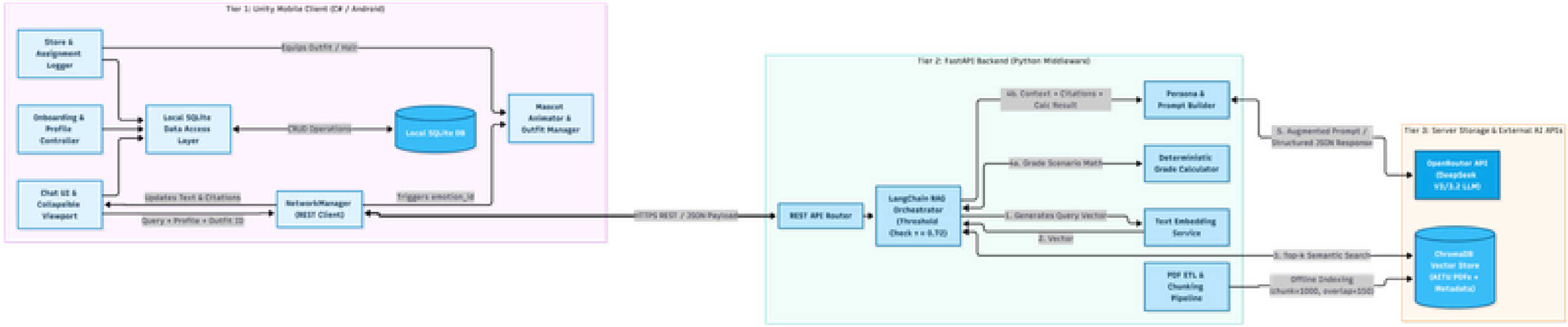
System Context Diagram (C4 Level 1) — External Actors (AITU Student, Admin), Central RemoodleChat System, and External OpenRouter API / AITU PDFs.



UML Use Case Diagram — Maps student interactions including Onboarding, Cited Q&A, Grade Calculation, Assignment Logging, and Store Outfit Persona Shifts.

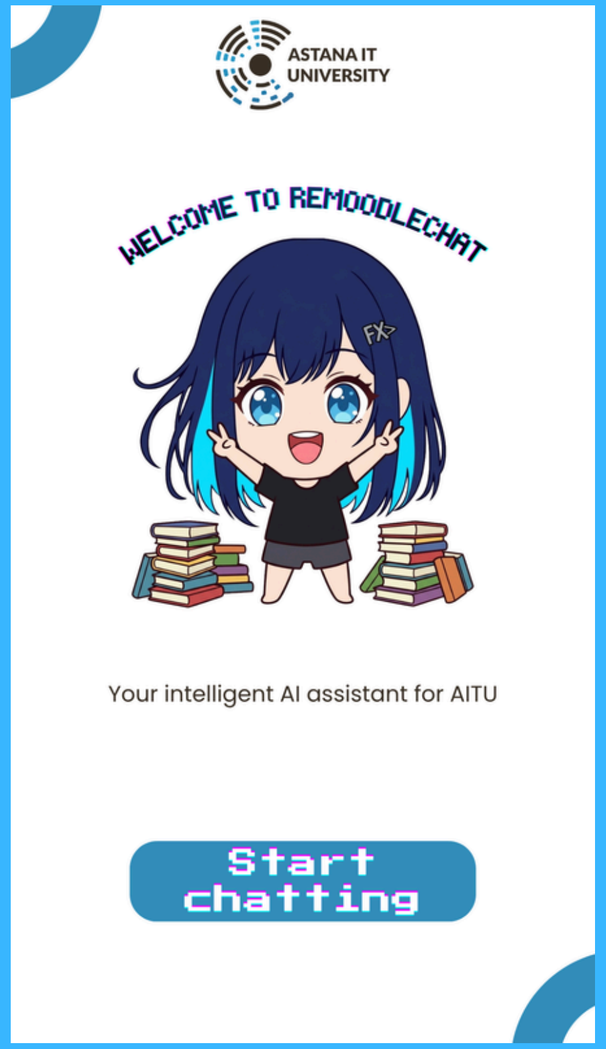


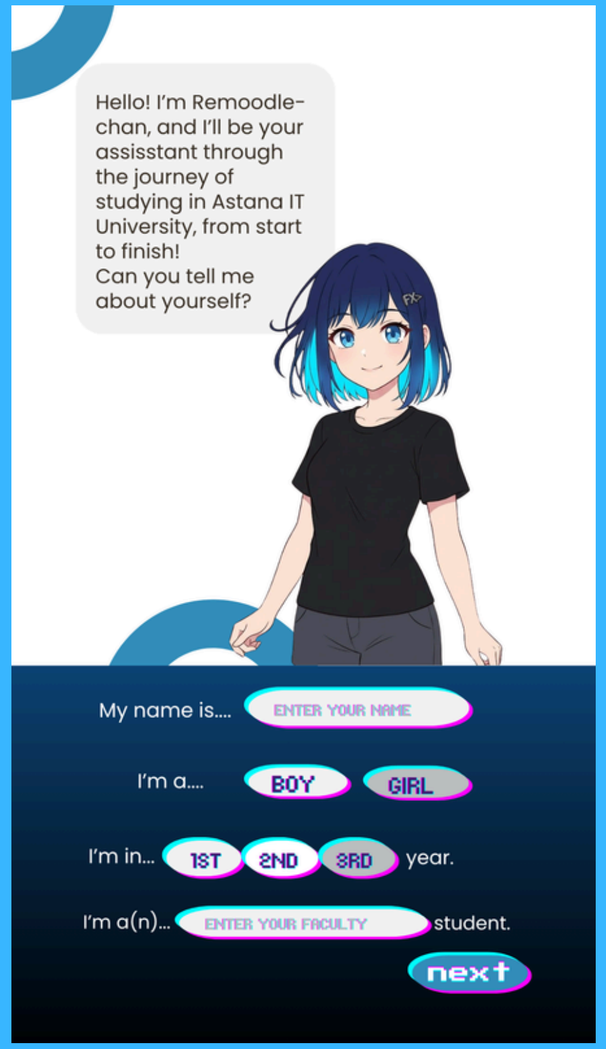
Architecture



- **Users**: user_id (PK), name, gender ('BOY'|'GIRL'), study_year (1..3), faculty, points_balance, equipped_hair_id, equipped_style_id, persona_shift_enabled (BOOL).
- **Assignments**: assignment_id (PK), user_id (FK), course_name, grade_score (0..100), points_awarded (1:1).
- **StoreItems**: item_id (PK), item_name, category ('HAIR'|'STYLE'), price_points, sprite_asset_key, persona_prompt_modifier.
- **UserInventory**: inventory_id (PK), user_id (FK), item_id (FK), purchased_at.
- **ChatMessages**: message_id (PK), session_id, sender, message_text, citation_json, emotion_id.

Prototype Design - Onboarding



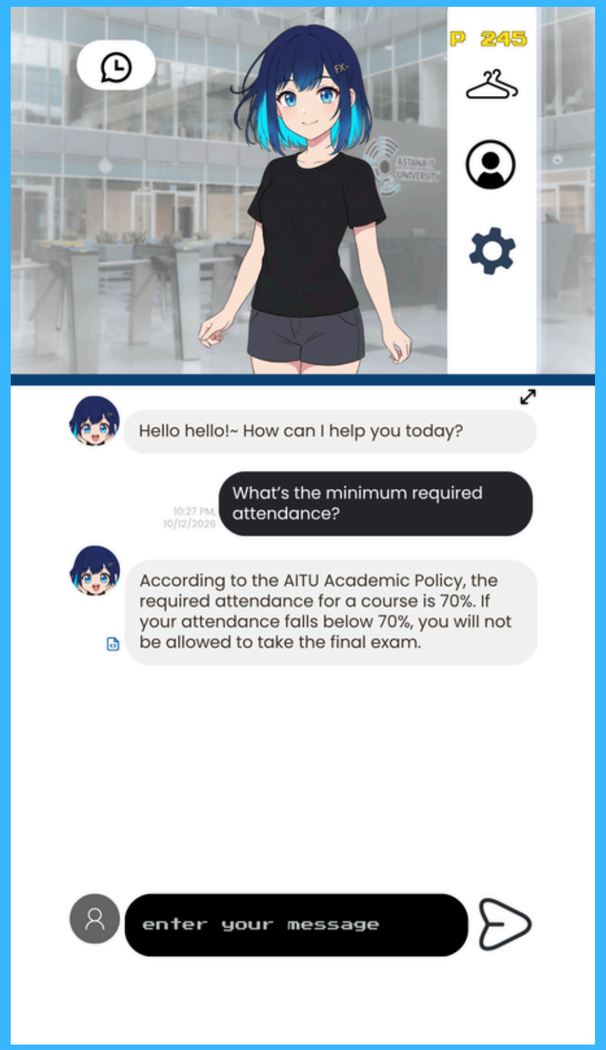


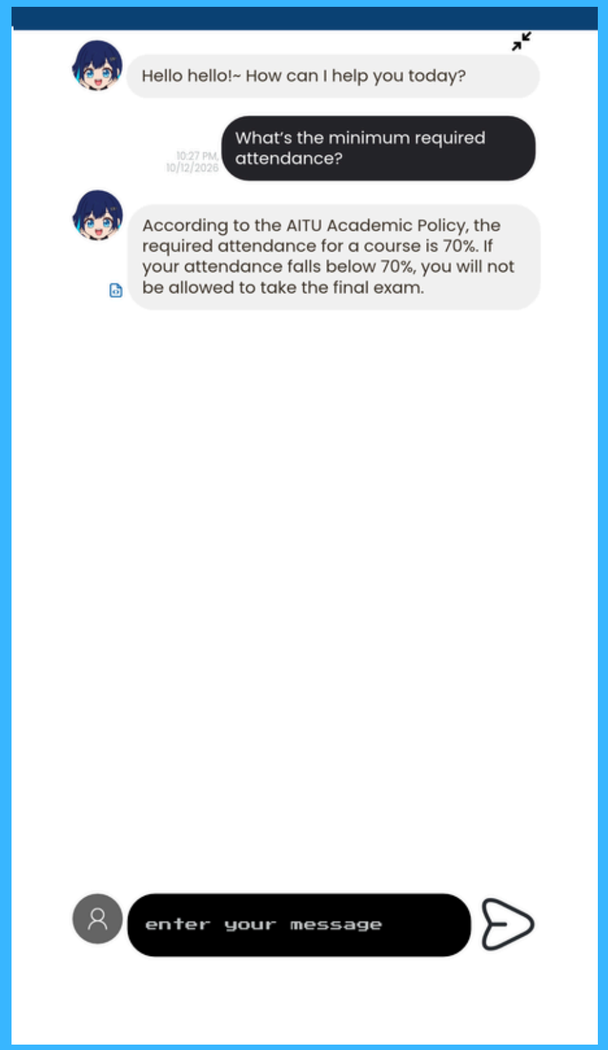
User starts with pressing “Start chatting” and providing relevant info for personalization

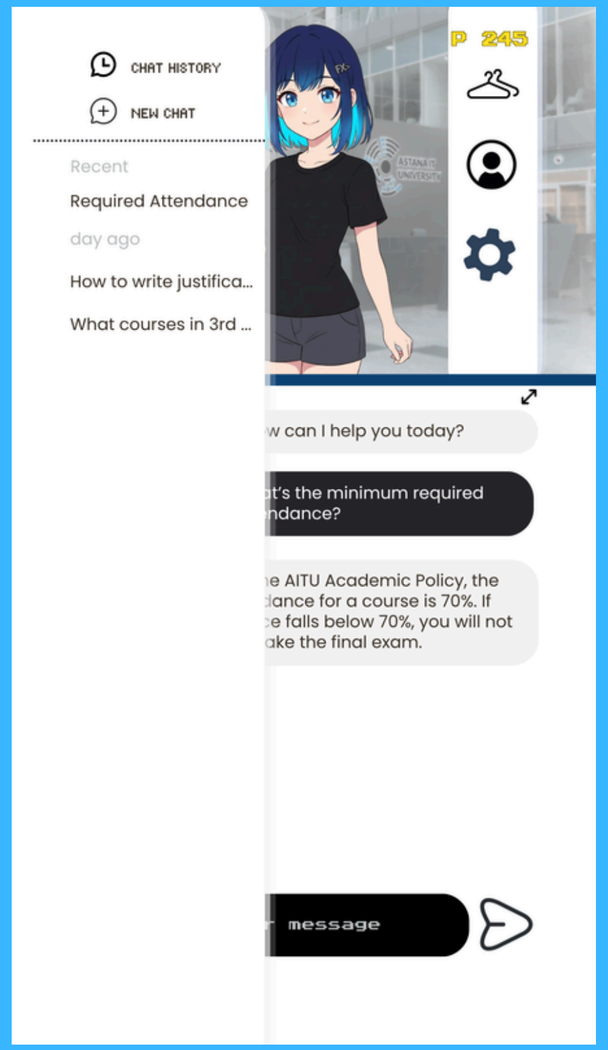
Design Features:

- Clean aesthetic utilizing AITU official colors (White, Blue, Charcoal).
- Clear value proposition welcoming the student.
- Frictionless entry point

Prototype Design - Chat







User accesses chat history by pressing on chat history icon on top left. On right side of the panel, there's a Profile Icon, Customization Icon and Settings Icon. User can hide the sprite and unnecessary panels by pressing the "enlarge" icon on the top right of the chat window.

Prototype Design - “Store” system







User can spend POINTS (P) in the store customizing the mascot by buying sprite modifications. POINTS are earned through completing and getting grades for assignments. Points obtained in assignments convert 1-1 to POINTS in-app. (LMS lacks API, so user has to manually add assignments and grades)

Verification & Evaluation Plan

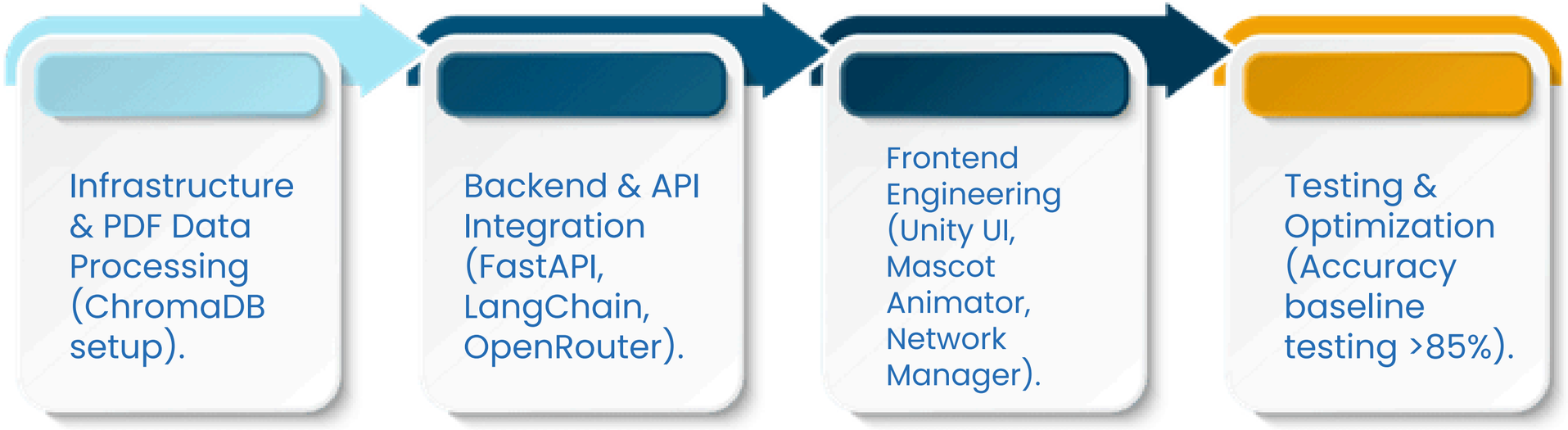
Benchmark Dataset (N = 70 Queries) & Test Conditions:

- **50 In-Scope Queries:** Complex multi-clause questions on AITU attendance, FX/retakes, GPA math, and syllabus deadlines paired with ground-truth answers and exact citations.
- **20 Out-of-Scope Queries:** Unindexed questions designed to test hallucination abstention (tau < 0.72).
- **Baseline Comparator:** Standard non-RAG LLM (ChatGPT-4o / DeepSeek V3 without RAG) tested on the identical 70 questions.

Evaluation Framework & Target Metrics (RAGAS + System Profiling):

- **Factual accuracy > 85% on the 50 in-scope queries.**
- **>= 95% fallback warning activation on the 20 out-of-scope queries.**
- **100% exact grade caculation accuracy via Python calculator tool.**
- **Response speed < 8.0 seconds from Unity request dispatch to UI rendering.**

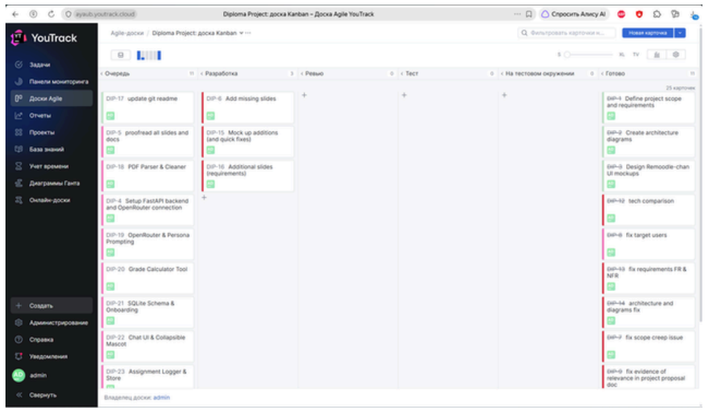
Plan for Future Work



Key Risks & Concrete Mitigations:

R-1 (API Latency > 8s): Mitigated via FastAPI async requests, local SQLite/Redis FAQ caching, and OpenRouter model fallback after 5s.
R-2 (PDF table corruption): Mitigated via layout-aware pdfplumber parsing into structured Markdown/JSON blocks.
R-3 (LLM math hallucinations): Mitigated by routing grade queries to deterministic calculate_required_final() Python function.
R-4 (Persona overriding facts): Mitigated by decoupling verbatim rule extraction from conversational persona styling in JSON output.

Supervisor’s feedback



Active YouTrack Board Screenshot

Checkpoint	Deliverable Reviewed	Supervisor Feedback (Zhanna Sadirmekova)	Engineering Decision & Action Taken
Checkpoint 1 (Late Sept 2026)	Initial Topic & Problem Formulation	Approved topic; recommended conducting user interviews with students to empirically validate relevance and derive requirements.	Conducted a 6-question interview study with 6 AITU students
Checkpoint 2 (Mid-Oct 2026)	Draft 1 of Presentation & Proposal	Prose/design needed improvement; missing GitHub/YouTrack links and UI mockups; 15s latency was too slow; needed stronger literature-to-requirement traceability.	Redesigned slides and UI mockups; tightened NFR-2 latency from 15s to <8s; added similarity threshold ($\tau = 0.72$).
Checkpoint 3 (Mid-Oct 2026)	Draft 2 (UI Prototype, Lit Review, Tech Table)	Approved progress; requested logical slide reordering, a Risk Management table, replacing 4 sources with different sources, and adding a local DB for user profiles.	Reordered slides; added Risk Matrix; replaced 4 URLs with 2 peer-reviewed researches and 2 technical documents; added SQLite for profiles/store.

Conclusion

- **User interviews and literature confirm that AITU students need an instant, verifiable alternative to 10–20 minute manual PDF searches on Moodle.**

- **Technically Sound:** Combining a FastAPI + LangChain + ChromaDB RAG backend (tau = 0.72 cutoff, deterministic grade math, <8s latency) with a Unity + SQLite frontend satisfies all 7 FRs and 4 NFRs.

- **Ready for Implementation:** Repository structure, YouTrack tasks (DIP-1 to DIP-25), C4/UML diagrams, UI prototypes, and the 70-query RAGAS verification plan are established and baselined under Git tag prob-period-1



References

- Rafique, R., et al. (2026). EduAssist: Enhancing Academic Skills and Engagement Through Personalization and Gamification. Pak. J. Eng. Technol.
- Ong, C. A., et al. (2026). Exploring Student Perceptions Towards AskNarelle: A Context-Specific LLM-Powered Educational Assistant. ACM.
- Shihab, S. M. (2026). AI-Enabled Higher Education Management. IJEETR.
- Niyazova, G. Z., et al. (2026). The effects of gamified AI-supported digital learning environments on personalized learning and student engagement. Frontiers in Education.
- Wu, W., et al. (2025). MultiRAG: A Knowledge-guided Framework for Mitigating Hallucination in Multi-source RAG. arXiv.
- Sanyal, P., et al. (2026). RAGMail: a cloud-based retrieval-augmented framework for reducing hallucinations in LLM text generation. Scientific Reports.
- Yuan, X., et al. (2026). SeaRAG: Reducing Hallucination in Retrieval-Augmented Generation via Statement-Entity Adaptive Ranking. ACM Web Conference.
- Joseph, A. (2025). Reducing Hallucinations in Large Language Models Through Integrated Self-Verification and Retrieval-Augmented Generation. ASME IDETC/CIE2025.
- Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2024). RAGAS: Automated Evaluation of Retrieval Augmented Generation. In Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL): System Demonstrations.
- Ryan, R. M., & Deci, E. L. (2020). Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. Contemporary Educational Psychology, 61, 101860.
- Chroma Technologies (2026).ChromaDB Technical Architecture: HNSW Vector Indexing, Metadata Filtering, and Persistent Local Storage. https://docs.trychroma.com.
- Unity Technologies (2026).Unity Mecanim Animation System: Animator Controllers, State Machines, and 2D Sprite Swap Pipelines. https://docs.unity3d.com.

Thank You!

Questions?

Presented By:

Ayauzhan Baurzhanova
| 3rd year |
Software Engineering